PH1201: Electricity and Magnetism

Tutorial Session-8

April 23, 2024

Problem 1: Solenoid

Find the magnetic field at point P on the axis of a tightly wound solenoid (helical coil) consisting of n turns per unit length wrapped around a cylindrical tube of radius a and carrying current I (Fig. 1). Express your answer in terms of θ_1 and θ_2 (it's easiest that way).

Use the result: The magnetic field a distance z above the center of a circular loop of radius R, which carries a steady current I is given by $\dfrac{\mu_0 I}{2}\dfrac{R^2}{(R^2+z^2)^{3/2}}$

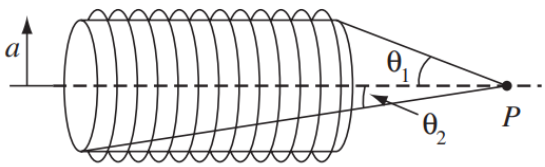


Figure 1

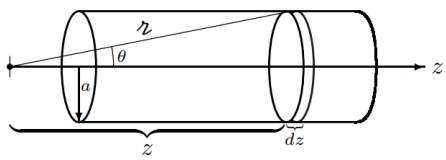
Problem 5.11

Use Eq. 5.41 for a ring of width dz, with $I \to nI\,dz$:

$B = \dfrac{\mu_0 nI}{2}\displaystyle\int \frac{a^2}{(a^2+z^2)^{3/2}}\,dz$. But $z = a\cot\theta$,

so $dz = -\dfrac{a}{\sin^2\theta}\,d\theta$, and $\dfrac{1}{(a^2+z^2)^{3/2}} = \dfrac{\sin^3\theta}{a^3}$.

So



$$B = \frac{\mu_0 nI}{2}\int \frac{a^2\sin^3\theta}{a^3\sin^2\theta}(-a\,d\theta) = -\frac{\mu_0 nI}{2}\int \sin\theta\,d\theta = \frac{\mu_0 nI}{2}\cos\theta\Big|_{\theta_1}^{\theta_2} = \boxed{\frac{\mu_0 nI}{2}(\cos\theta_2 - \cos\theta_1).}$$

For an infinite solenoid, $\theta_2 = 0$, $\theta_1 = \pi$, so $(\cos\theta_2 - \cos\theta_1) = 1 - (-1) = 2$, and $B = \boxed{\mu_0 nI.}$ ✓

Figure 2: Solution

Alternate of Problem 1

Problem On Finding Magnetic Field (Application Of Biot-Savart Law): Current I flows through an infinitely long straight wire having the cross-section of the form of a very thin half-ring of radius R (the diagram may be helpful). Find the magnetic field at any point on the axis of the wire (or equivalently at the centre of half-ring cross-section).
Answer: $\mu_0 \frac{I}{\pi^2 R}$

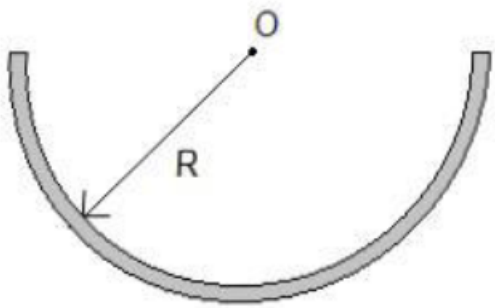


Figure 3

Problem 2: Vector potential for a wire

We know $\mathbf{E} = -\boldsymbol{\nabla} V$ and $\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}$, we call $\mathbf{A}$, *vector potential.*
Consider a long straight wire carrying a current I (coming toward us out of the page) flowing along the positive z axis. Outside the wire, what is the **vector** potential $\mathbf{A}$?
(HW: Rewrite $\mathbf{B}$ and $\mathbf{A}$ in terms of Cartesian coordinates, and verify that $\boldsymbol{\nabla} \times \mathbf{A} = \mathbf{B}$.)

Solution:
We know what the magnetic field of the straight wire looks like.

$$\mathbf{B} = \frac{\mu_0 I}{2\pi r}\hat{\theta}$$

We want to find a vector field $\mathbf{A}$ whose curl equals this $\mathbf{B}$.
The expression for the curl in cylindrical coordinates

$$\left(\frac{1}{r}\frac{\partial A_z}{\partial \theta} - \frac{\partial A_\theta}{\partial z}\right)\hat{\boldsymbol{r}} + \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right)\hat{\boldsymbol{\theta}} + \frac{1}{r}\left(\frac{\partial\left(rA_\theta\right)}{\partial r} - \frac{\partial A_r}{\partial \theta}\right)\hat{\mathbf{z}}$$

we are concerned only with the $\hat{\theta}$ component of the curl expression. Due to the symmetry along the z axis, we can't have any z dependence, so we are left with only the θ term.

$$\boldsymbol{\nabla} \times \mathbf{A} = \mathbf{B} \implies -\frac{\partial A_z}{\partial r} = \frac{\mu_0 I}{2\pi r} \implies \mathbf{A} = -\hat{z}\frac{\mu_0 I}{2\pi}\ln r$$

Alternate problem 2

Problem On Lorentz Force:

Uniform magnetic field (B) exists in a region of circular cross-section with radius R, the magnetic field is normal to the cross-section (as shown in the figure). A charged particle (mass M) of charge q enters into the region heading towards the centre of the circular cross-section with a speed v. The particle, being deflected by magnetic field, leaves the circular region. Find the time taken by the particle to leave off the region of magnetic field. [$Answer$: $\frac{2M}{qB}\tan^{-1}(\frac{qBR}{Mv})$]]

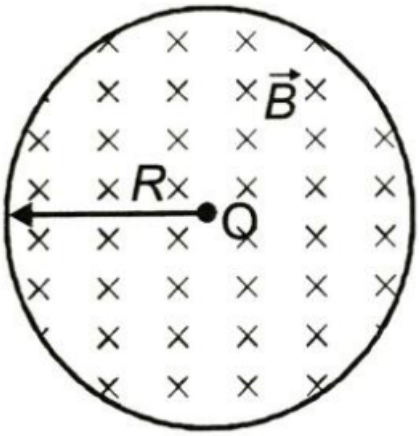


Figure 4

Problem 3: Graded

A current I runs along an arbitrarily shaped wire that connects two given points, as shown in Fig. 2 (it need not lie in a plane). Show that the magnetic field at distant locations is essentially the same as the field due to a straight wire with current I running between the two points.

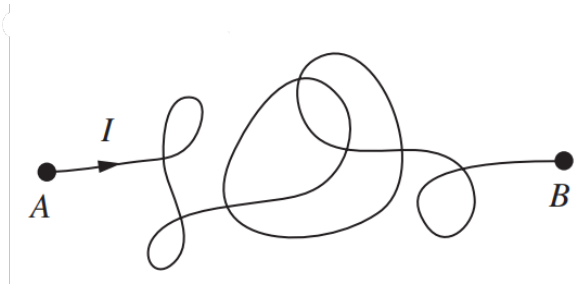


Figure 5

Alternate of Problem 3:

Final Problem: The hollow disk is formed by a tightly wound coil of thin wires such than the disk carries a current I across it's radial cross section. The inner and outer radii of the disk are r_1 and r_2. Find the magnetic field intensity (only magnitude is sufficient) at an axial point situated at a distance x from centre of the coil.